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BEGINNER-FRIENDLY GUIDE

Docker Explained — From Zero to Deploy

→ Swipe through all 6 slides

Shubham Bhati
linkedin.com/in/bhatishubham

What Problem Docker Solves

It works on MY machine...

```
Without Docker:
  Dev uses Java 17, MySQL 8
  Server has Java 11, MySQL 5
  → App crashes in production

With Docker:
  Package app + ALL dependencies
  → Same everywhere
  → Dev = Staging = Production
```

💡 Docker = ship your entire environment

Shubham Bhati
linkedin.com/in/bhatishubham

Image vs Container

Blueprint vs Running instance

```
Image:
  → Read-only template
  → Like a Java class
  → Built from Dockerfile

Container:
  → Running instance of image
  → Like a Java object
  → Has its own process, network, storage

One image → many containers
```

💡 Image = recipe. Container = cooked dish.

Dockerfile — Your Recipe

5 lines to containerize Spring Boot

```
FROM openjdk:17-slim
WORKDIR /app
COPY target/app.jar app.jar
EXPOSE 8080
CMD ["java", "-jar", "app.jar"]

→ Start from Java 17 base
→ Copy your JAR file
→ Tell Docker which port
→ Run the app
```

💡 This is all you need for Spring Boot

Docker Compose — Multi-Service

App + DB + Cache in one command

```
services:
  app:
    build: .
    ports: ["8080:8080"]
    depends_on: [db, redis]
  db:
    image: mysql:8
    environment:
      MYSQL_ROOT_PASSWORD: secret
  redis:
    image: redis:7

→ docker-compose up
→ Everything starts together
```

💡 docker-compose up = entire system running

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Essential Commands

The only 8 you need

```
docker build -t myapp .
docker run -p 8080:8080 myapp
docker ps          # running containers
docker logs    # see output
docker stop    # stop container
docker images      # list images
docker-compose up  # start all
docker-compose down # stop all
```

💡 Master these 8. That covers 95% of daily use.

Shubham Bhati
linkedin.com/in/bhatishubham

Docker in Interviews

Questions they'll ask

```
Q: Image vs Container?
A: Blueprint vs running instance

Q: How to reduce image size?
A: Multi-stage builds + slim base

Q: How services communicate?
A: Docker network + service names

Q: Data persistence?
A: Volumes (docker volume create)
```

💡 Docker is asked in every backend interview now

Shubham Bhati
linkedin.com/in/bhatishubham

```
done = true
```

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Shubham Bhati
linkedin.com/in/bhatishubham